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MSc Computer Science researcher with training in economics (CERGE-EI).
Research interests include artificial intelligence, deep learning, and quantitative modelling applied to economic problems. Experience at the European Central Bank, machine learning research and in software engineering.

Education

08/2025–04/2026 MS Economic Research, CERGE-EI, Prague, Czech Republic
- Postgraduate training coursework in Advanced Microeconomics, Statistics, and Macroeconomics.

10/2019–02/2022 MS Computer Science, Faculty of Electrical Engineering and Computing, Zagreb, Croatia
- Thesis topic: Semantic segmentation of buildings in satellite imagery
- Recommendation letter from prof. dr. sc. Siniša Šegvić
- Additional activities: Football sports section;; English language Courses; Communication skills

10/2016–7/2019 BS Computer Science, Faculty of Electrical Engineering and Computing, Zagreb, Croatia
- Thesis topic: Convolutional Models for Person Localization in Sports Broadcast
- Additional activities: Haskell, Java Programming Language, Linux, Arduino and Raspberry PI workshops, Discrete & Continuous Mathematics; CubeSat Satellite Section

Skills and Relevant Coursework

- Technologies and Languages: Java, Python, SQL, R, Linux, PyTorch, Excel
- Skills: Writing and Communication; System Modelling; Presentation; Engineering mindset; Research
- Relevant Coursework: Algorithms and data structures; Machine learning; Text analysis and retrieval; Deep learning; Financial mathematics; Stochastic processes; Fuzzy, evolutionary and neuro computing; Discrete & Continuous Mathematics; Software Design; Statistical Data Analysis; Operational Research

Industry Experience

09/2026 - present, AI Researcher, Faculty of Electrical Engineering and Computing, Croatia, Zagreb
- Description: Researched and prototyped generative deep learning methods for automated ship blueprint design. Designed the solution architecture for 3D generation and presented a literature survey on models and benchmarks.
- Skills: PyTorch, Python, Scientific literature surveys, Scientific proofreading and experimentation

01/2024-07/2025, Entrepreneur, Stipic Construction company, Croatia, Zagreb
- *Description*: Founded and ran a construction company's digital operations, building web and Android solutions. Delivered employee GPS tracking, employee database, URL analytics, and an inflation-adjusted accounting tracker with bank API integration. Applied data science and operational research to business problems, including a Poisson-based system that predicts incoming messages to prioritize attention and a transportation-problem model that allocates resources to projects.
- *Skills*: Python, Databases, Linux, Android, Spreadsheet, Git, Google Cloud & Digital Ocean management

1/2023-8/2023 Traineeship in Directorate General Market Operations, European Central Bank, Frankfurt am Main, Germany
- *Description*: Central bank analyst specializing in money markets, excess liquidity, and monetary policy implementation. Prepared materials for Eurosystem decision-making bodies and supported TLTRO operations and counterparty monitoring. Built Python analytical tools, a reserve requirement penalty application, and introduced Dagster with CI/CD to the department.
- *Skills*: Python, Oracle Database, Tableau, Excel, Bloomberg, Dagster framework, Essentials of Monetary policy and Economy, Presentation skills, Data analysis, Research.

3/2022-9/2022 Software Engineer, Amphinicy Technologies, Zagreb, Croatia
- *Description* : Telecommunications software engineer specializing in Java EE, MVC systems, event-driven architectures, and secure authentication. Built a system-wide telemetry solution (Composite pattern) for the latest product release.
- *Skills:* Java EE, Docker-compose, IntelliJ, Git, Software design and implementation, Managing large code base, technical documentation

3/2021-9/2021 ML engineer, PhishAR, Zagreb, Croatia
- *Description*: Designed and built a modular, Dockerized TensorFlow data pipeline that prepared mobile-captured images from Google Cloud Storage for training on-device computer vision deep learning models. I used FFmpeg for frame extraction, Tesseract OCR for text detection, and OpenCV perspective warping to correct skewed photos;
- *Skills*: Python, Tensorflow, Git, Docker, Google Cloud Platform, Amazon Mechanical Turk

7/2019-9/2019 Student internship in data science/engineering, CROZ, Zagreb, Croatia
- *Description:* Reduced picker walking distance by 17% by building a Spark-based warehouse optimization algorithm over 30 GB of historical orders; designed the supporting Oracle relational data model and database for warehouse inventory and order data.
- Skills: Python, Apache Spark, Excel, Oracle SQL

Academy Experience

- 2020 Researcher, Text Analysis and Retrieval Course, DSAA 2020 conference
  - Name: „Are you Human? Detecting Bots on Twitter Using Bert" (IEEE xplore and ResearchGate)
  - Model for text classification using BERT, feature exploration, exploratory data analysis of tweets
  - Presenting work on the online conference [2].
- 2020 University days event, 2020 [3] : *Presenting paper to future students and answering student questions.*
- 2019 Research assistant, FERsat project: *The aim of the project was monitoring of light pollution from space.*

[2] DSAA conference, Day 2 - http://dsaa2020.dsaa.co/technical-program
[3] Smotra sveučilišta - https://smotra.fer.unizg.hr/?p=215